

Practical File
of
Operating System and Linux Fundamentals
25CSE0104

Submitted
in partial fulfillment for the award of the degree

of
BACHELOR OF ENGINEERING

in
COMPUTER SCIENCE & ENGINEERING



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S.No.	Practical Name	Page No.	Date	Teacher Signature
1	a) Installation: Configuration & Customisations of Linux b) Introduction to GCC compiler: Basics of GCC, Compilation of program, Execution of program c) Time stamping, d) Automating the execution using makefile.	1-2	27/01/2026	
2	Basic Commands of Linux OS: Navigation (ls, cd), File manipulation (cp, mv, rm), and Viewing (cat, less), where is, what is and their options	3-4	03/02/2026	
3	Execute Directory-oriented commands: cd, pwd, mkdir, rmdir and their options	5-6	10/02/2026	
4	Execute the basic and user status commands like: su, sudo, man, help, history, who, whoami, id, uname, uptime, free, tty, cal, date, hostname, reboot, clear and their options	7-9	17/02/2026	
5	Write programs to create and manage processes using system calls such as fork(), exec(), and wait(). Study parent-child relationships and process lifecycle behaviour in Linux.	10-11	24/02/2026	
6	Write a program to implement Process concepts by printing the process ID.	12	10/03/2026	
7	Implement FCFS, SJF scheduling algorithms.	13-14	24/03/2026	
8	Implement Linux memory management commands: Overview, free, vmstat, Memory Allocation System Calls (malloc, sbrk). Analyse how memory is allocated and utilized by running processes.	15-17	31/03/2026	
9	Implement deadlock detection in C. (Banker's Algorithm)	18-20	07/04/2026	
10	File System Concepts: File attributes, access methods, directory structures, protection.	21-22	21/04/2026	
11	Execute File system commands: Comparing Files using diff, cmp, comm and their options	23-24	28/04/2026	
12	Execute File system <i>system-calls</i> : open, read, write, close, lseek, stat and their options	25-26	09/05/2026	
13	Execute Linux file management commands: chmod, chown, ln, mount, df, du and their options	27-28	12/05/2026	